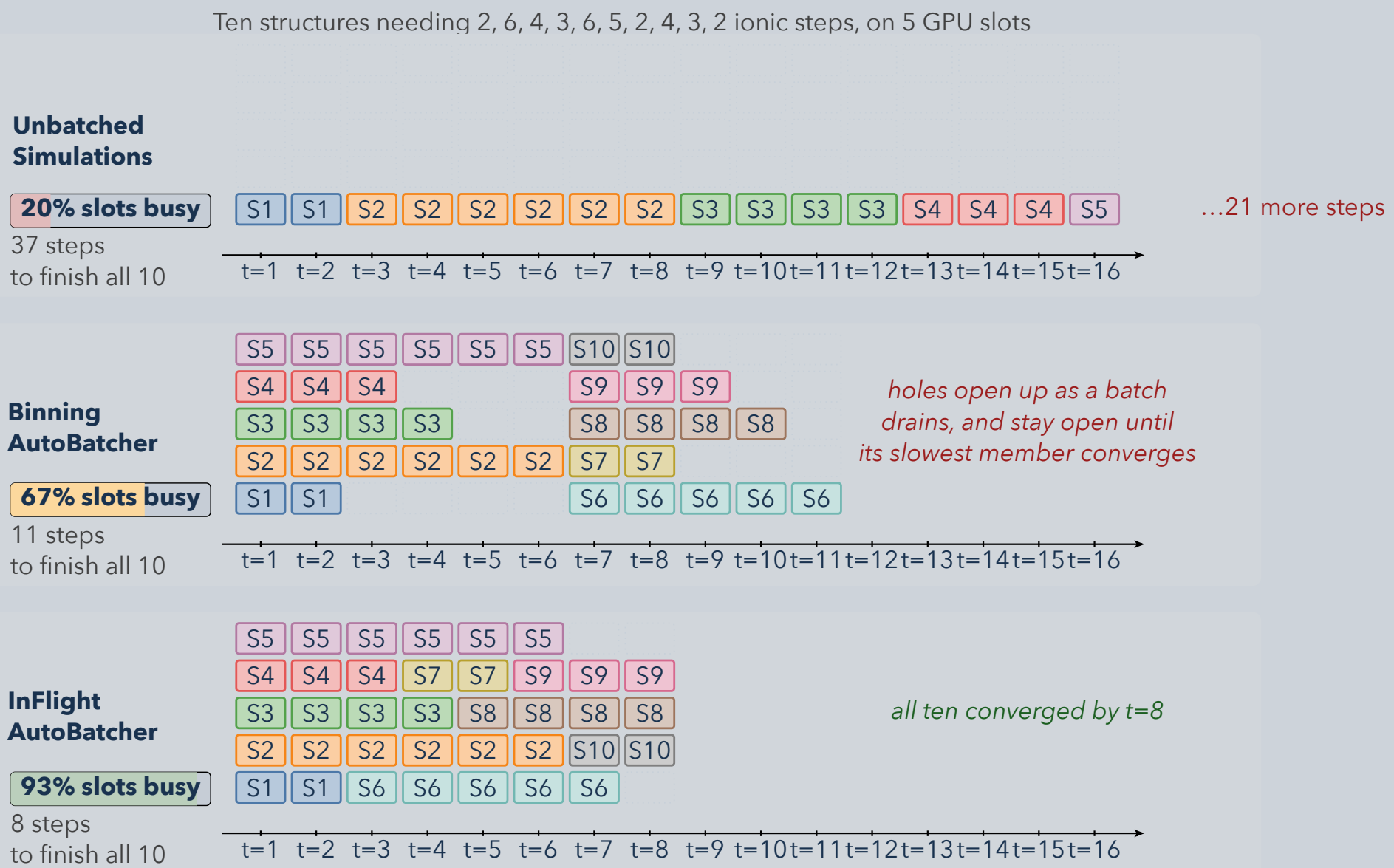
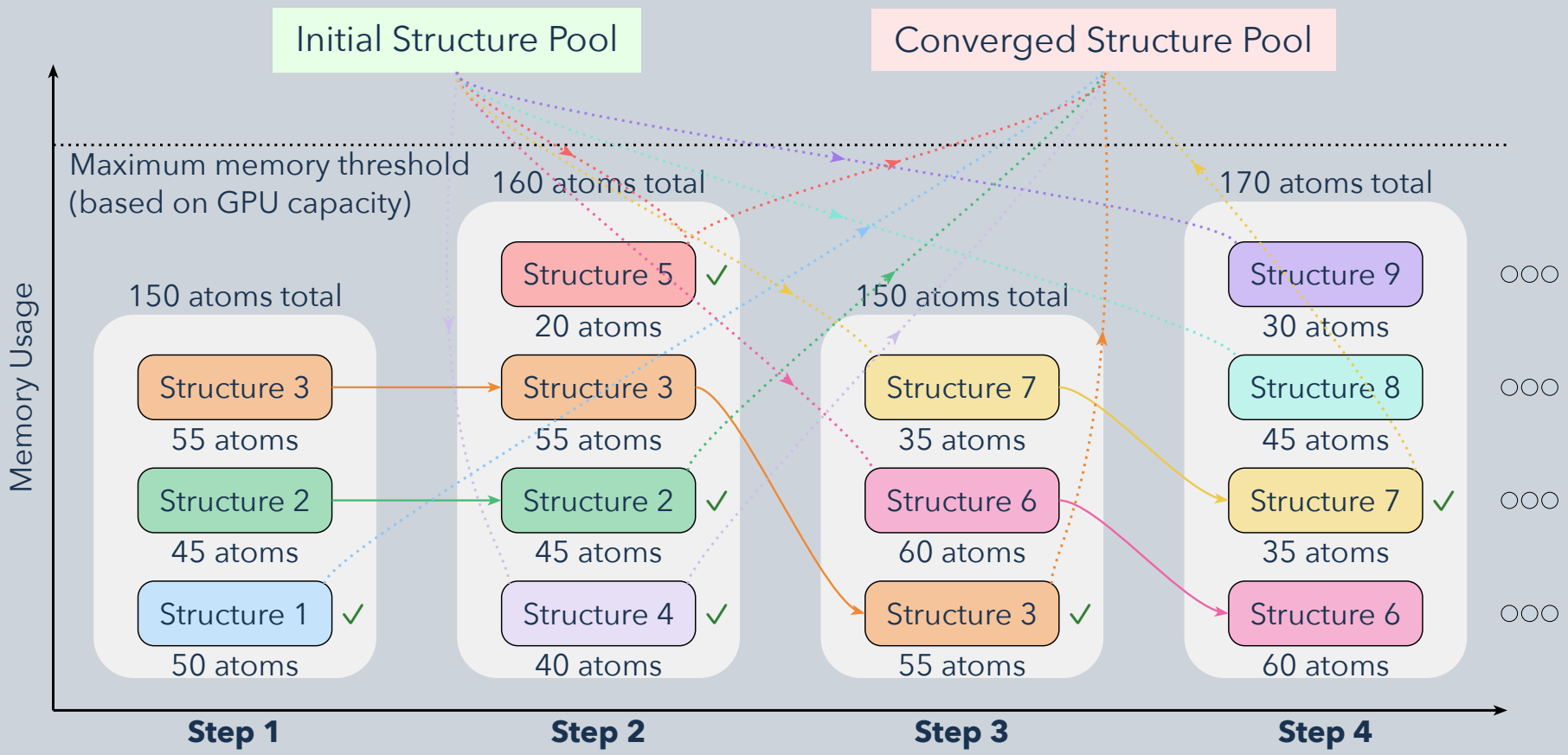
Small atomistic jobs can leave a GPU underused. Batching runs several together; replacing completed jobs keeps useful work in the batch.

1 Compare scheduling strategies



Unbatched leaves four of five slots idle. **Binning** holds each slot until the batch's slowest member converges. **In-flight** immediately reuses freed slots: a color change within a row marks a new structure taking over.

2 Follow a replacement



Each structure has its own convergence state. After a structure finishes, move it to the converged pool and fill the freed capacity from the initial pool.

Batch by resource use, not just job count. Atom counts, neighbor counts, memory, and model cost constrain admission. Utilization percentages in the schematic are illustrative, not measured benchmarks.